

Quiz-5

assumptions:

- (1) Steady flow
- (2) Incompressible flow; $\rho = \text{constant}$.
- (3) No flow or variation of properties in the z-direction
 $w = 0$ and $\frac{\partial}{\partial z} = 0$.
- (4) Fully developed flow, u, v, w
2D flow; consider flow in y-direction only.
i.e. ' v ' velocity component.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \text{--- (5)}$$

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = \rho g_x - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = \rho g_y - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right)$$

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = \rho g_z - \frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)$$

$$\rho g - \frac{\partial p}{\partial y} + \mu \frac{\partial^2 v}{\partial x^2} = 0$$

$$\mu \frac{\partial^2 v}{\partial x^2} - \frac{\partial p}{\partial y} + \rho g = 0$$

$$\mu \frac{\partial^2 v}{\partial x^2} - \frac{\partial p}{\partial y} - \rho g = 0$$

$$\mu \frac{\partial^2 v}{\partial x^2} = \left(\frac{\partial p}{\partial y} + \rho g \right)$$

integrate

$$\frac{\partial v}{\partial x} = \frac{x}{\mu} \left(\frac{\partial p}{\partial y} + \rho g \right) + C_1$$

At minimum velocity condition: $\frac{\partial v}{\partial x} = 0$ at $x = 0$

$\therefore C_1 = 0$

$$\frac{\partial v}{\partial x} = \frac{x}{\mu} \left(\frac{\partial p}{\partial y} + \rho g \right) = \frac{x \rho g}{\mu} \quad \left(\text{Neglect } \frac{\partial p}{\partial y} \right)$$

Surprise Quiz 5(open Book)

25/09/13

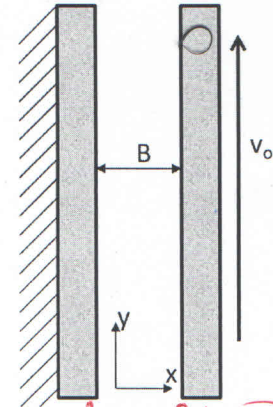
15 min, 10 points

NAME: *Solution*

ID:

Section:

Q1: A Newtonian fluid is confined between two broad parallel vertical plates, separated by a distance B , as shown in the figure to the right. The plate on the left is stationary; that the right plate is moving vertically upward at a constant velocity v_0 . Assume laminar flow at steady state. Now the velocity v_0 must be large enough to counteract the effect of gravity, otherwise some fluid will flow downward. In a particular system the distance between plates is 1 mm, fluid is an oil with density 900 kg/m^3 , and viscosity of 50 mPa.s . The pressure drop is negligible compared to ρg . What is the minimum upward velocity of the moving plate so all fluid move upwards.



- 2D flow, but consider y -directional flow only
- Steady state
- incompressible fluid; Continuity Eq.

So the left eq from N-S set of eqs is

$$\mu \frac{d^2 v}{dx^2} - \frac{dp}{dy} - \rho g = 0$$

Integration

$$\frac{dv}{dx} - \frac{x}{\mu} \left(\frac{dp}{dy} + \rho g \right) = C$$

At minimum velocity condition $\frac{dv}{dx} = 0$ at $x=0 \therefore C=0$

$$\frac{dv}{dx} = \frac{x \rho g}{\mu}$$

$$v = \frac{x^2 \rho g}{2\mu}$$

$$v_0 = \frac{B^2 \rho g}{2\mu} = \frac{(0.001)^2 \text{ m}^2 \times 900 \text{ kg/m}^3 \times 9.8 \text{ m/s}^2}{2 \times 0.05 \text{ kg/m.s}}$$

$$= 0.088 \text{ m/s}$$

$$v_0 = 88.2 \text{ mm/s}$$

$$\begin{aligned} 50 \text{ mPa.s} \\ = 10^{-3} \frac{\text{kg}}{\text{s.m}} \end{aligned}$$

Neglect $\frac{dp}{dy}$
integrating bet $x=0$ to B
 $v=0$ to v_0